DAY 3 OF 30

Work-spaces & capacity

How Fabric is organised.
And how the bill actually works.

F SKUs | Capacity Units | Throttling

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

THE PROBLEM

"Why is my report slow?" has a billing answer.

1	**Old world**	Each service had its own meter
2	**Fabric**	One meter for everything
3	**Result**	Workloads compete for it

If nobody understands the meter, **the first heavy job teaches everyone at once**.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

THE SIMPLE VERSION

Rooms in a building, one power meter.

A **workspace** is a room. A team keeps its lakehouses, pipelines and reports in it.

A **capacity** is the electricity. Every room plugged into it shares the same supply.

✓ **Workspace**	Where items live
✓ **Capacity**	The compute you pay for
✓ **The link**	Each workspace uses one capacity

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

HOW IT WORKS

How the meter works

1. Compute is counted in **Capacity Units**
2. You buy a size: **F2 up to F8192**
3. **Pay as you go** can be paused. **Reserved** costs less
4. **Bursting** lets a job run faster than the size allows
5. **Smoothing** spreads that usage over time

Throttling
Sustained overuse first delays interactive work, then rejects it, and finally rejects background jobs too.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

FOR ENGINEERS

Details worth knowing early

Interactive and background are smoothed differently

A report click is smoothed over minutes. A scheduled job is spread across 24 hours.

F64 is a licensing line

From F64 up, people with a free licence and the Viewer role can open Power BI content. Below it, viewers need Pro.

The Capacity Metrics app is the source of truth

It shows which item used how much, and when throttling started.

Overage is now an option

Capacity overage bills spikes at a premium rate instead of throttling. It is on by default for new F capacities.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

THE HONEST PART

One capacity or several?

ONE SHARED CAPACITY

Simple to run and to explain

Idle compute is shared, not wasted

Fine for a small or new team

SPLIT CAPACITIES

Production reports need protection

Dev experiments are unpredictable

Departments want separate bills

> **The catch:** a Spark experiment in a dev workspace and the CEO's dashboard can share one meter. Separate them before it hurts, not after.

Keshav Maheshwari

Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

WATCH OUT

Four mistakes I would avoid

1. **Sizing from a guess.** Start small, measure, then resize
2. **Dev and prod on one capacity** with no guardrails
3. **Nobody owns the Metrics app**, so throttling is a surprise
4. **Leaving pay-as-you-go running** on nights and weekends

Capacity is a design decision, **not a purchase order**.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Swipe →

SAVE THIS

Capacity in five lines

1. **Workspace** is where things live
2. **Capacity** is the compute they run on
3. **Everything shares it**, so plan for contention
4. **Smoothing helps**, throttling is the limit
5. **Measure first**, then choose the size

Tomorrow: The Lakehouse.
The item most Fabric projects start with.

Keshav Maheshwari
Data Engineer | Databricks · AWS & Azure · Streaming · Applied AI

Follow for Day 4